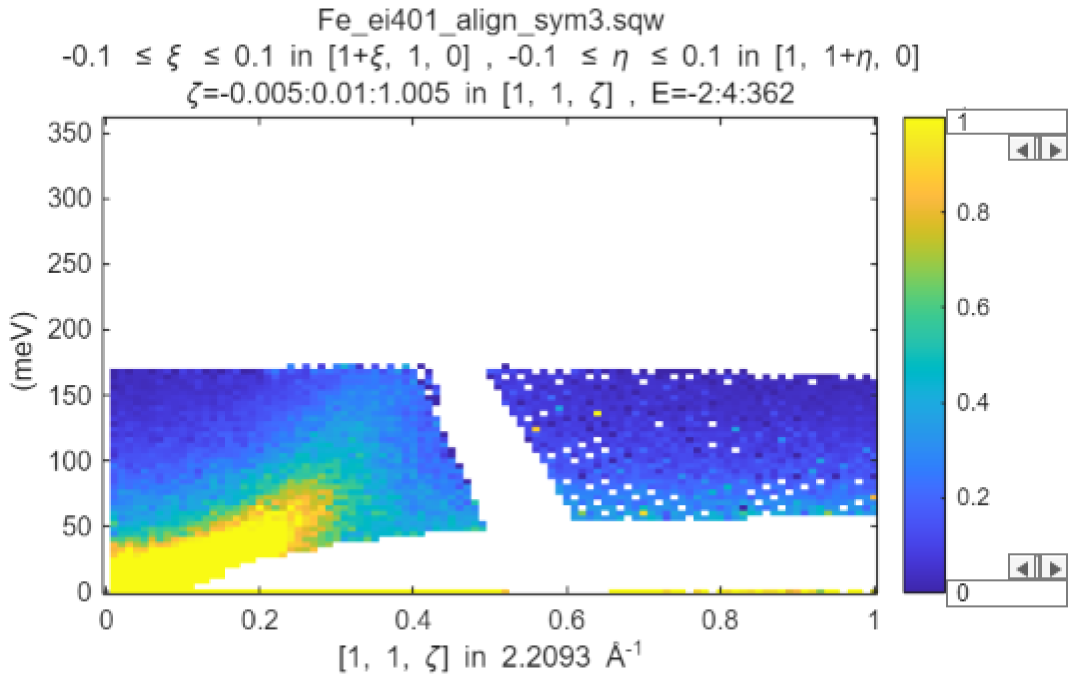
```
root_path = fileparts(mfilename("fullpath"));
root_path  = pwd;
data_path = fullfile(root_path,'sym4D_cutsAndFits');

if ~exist('cuts2fit400','var')
    ld = load(fullfile(data_path,'multicuts_fit_dataGH_ei400meV.mat'));
    cuts2fit400 = ld.cuts2fit400;
end
if exist('used_ds400','var')
    used_ds    = used_ds400;
else
    used_ds = containers.Map()
    sel_info = struct('cuts',[],'en_range',[],'fname','','bg_par',[]);
    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(1);
    sel_info.en_range = 50:10:160;
    sel_info.cut_dim  = 2;
    sel_info.fname='EnFit_Ei400ref001off110_2D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei400Br110dir001')=sel_info;
    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(2);
    sel_info.en_range = 50:10:200;
    sel_info.cut_dim  = 2;
    sel_info.fname='EnFit_Ei400ref001off200_2D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei400Br200dir001')=sel_info;

    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(1);
    sel_info.en_range = 50:10:160;
    sel_info.cut_dim  = 1;
    sel_info.fname='EnFit_Ei400ref001off110_1D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei400Br110dir001_1D')=sel_info;
    %
    sel_info.cuts = cuts2fit400.cuts_dir001FFfit(2);
    sel_info.en_range = 50:10:200;
    sel_info.cut_dim  = 1;
    sel_info.fname='EnFit_Ei400ref001off200_1D_constFFCut';
    sel_info.bg_par= {};
    used_ds('Ei400Br200dir001_1D')=sel_info;
    used_ds400 = used_ds;
end
```

```
used_ds = 

  Map with properties:

        Count: 0
      KeyType: char
    ValueType: any

  Show all accessible properties of Map
```

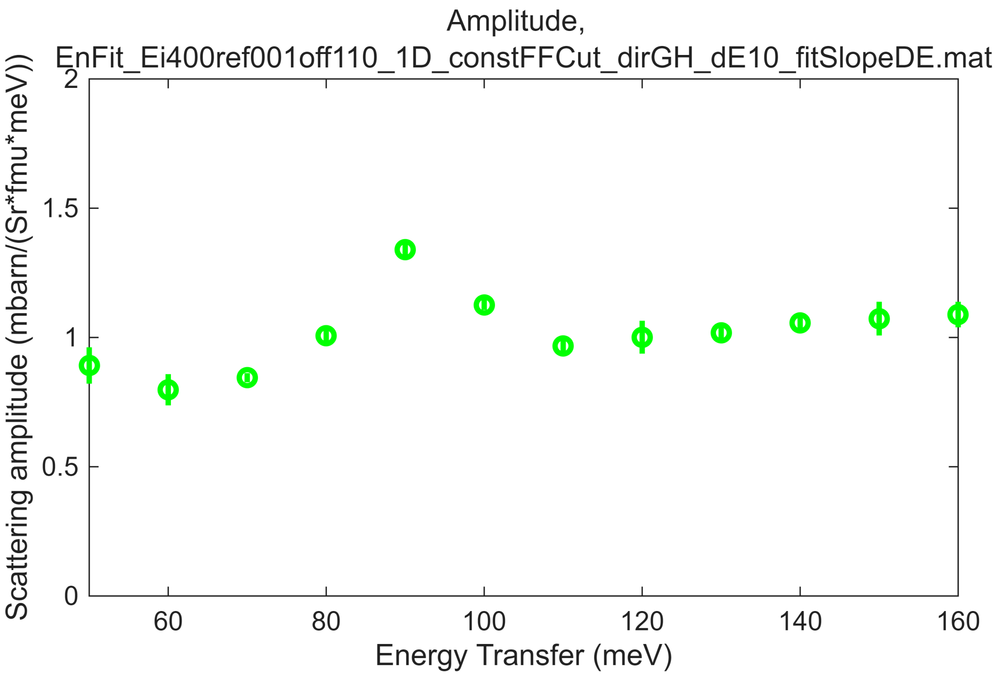
```
%cuts_list = cuts2fit400.cutsS_list; % all available cuts
%cuts_list = {cuts2fit400.cut_b200_200sel}; % cuts in low ff direction arount
<200>
%cuts_list = cuts2fit400.cuts_dir001FFfit(1);
select= 'Ei400Br110dir001_1D';
sel_info = used_ds(select);

cuts_list = sel_info.cuts;

mi = maps_instrument(400,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for i=1:numel(cuts_list)
    cuts_list{i} = cuts_list{i}.set_instrument(mi);
    cuts_list{i} = cuts_list{i}.set_sample(sample);
    plot(cuts_list{i}); lz 0 1; keep_figure;
end
```

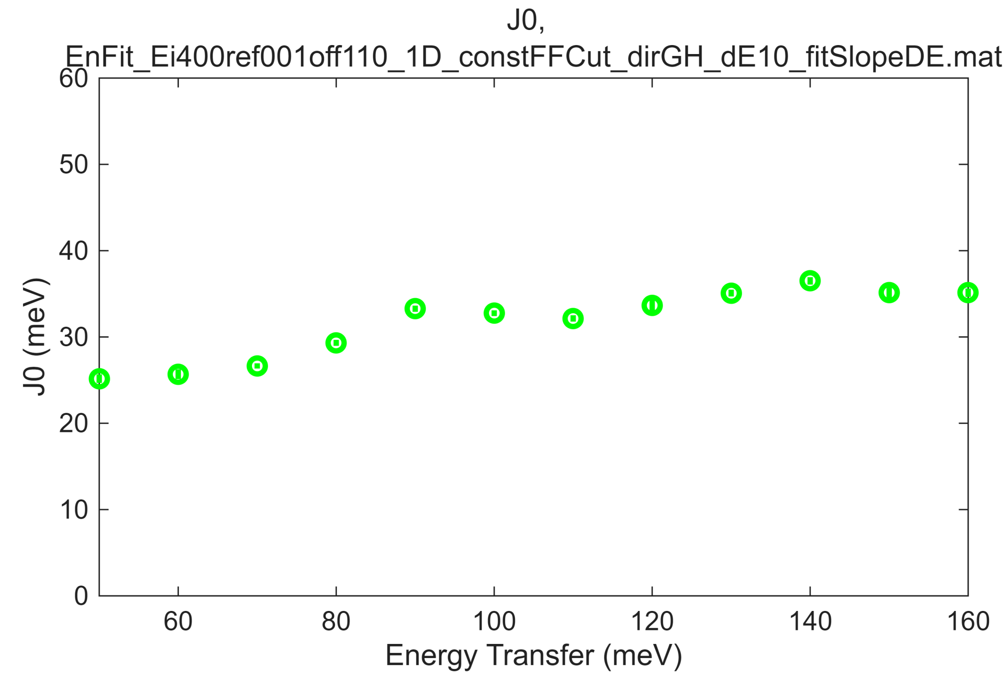


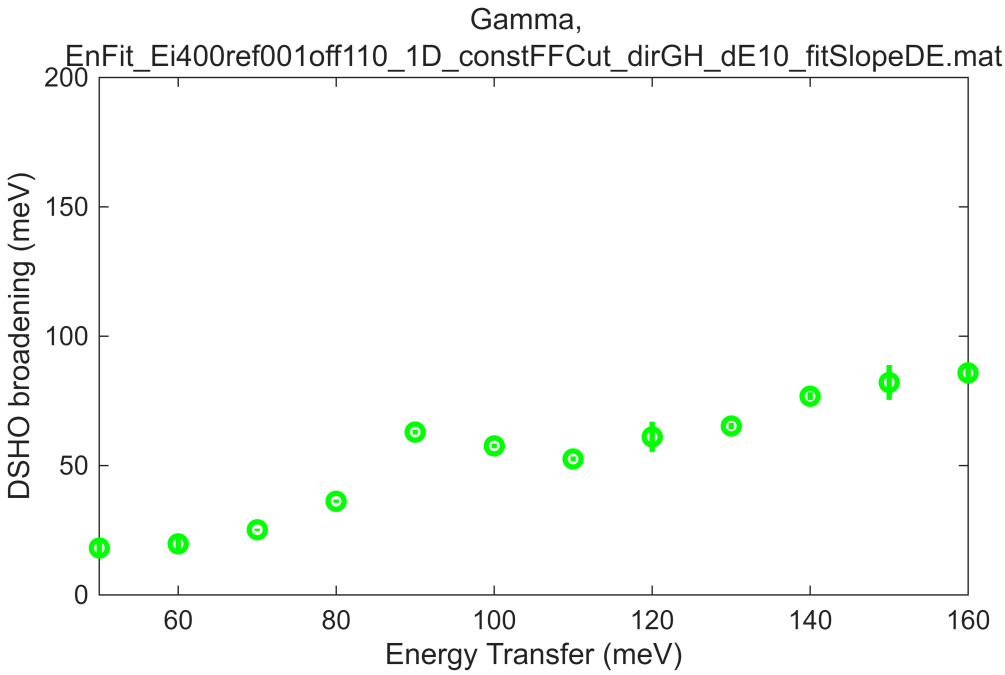

```
dir_name  = "GH";

dE_step = 4; %original energy transfer step data were binned to. No point in
going finer
half_dE = 5; % half width of data binning
%cut_en = 50:10:260;
%cut_en = 50:10:160; %cuts2fit400.cuts_dir001FFfit(1);
cut_en = sel_info.en_range; %cuts2fit400.cuts_dir001FFfit(2);
cut_dimensions = sel_info.cut_dim;

%cl= {cuts_list{3}};
%cl = {w2_test_cut};
% if isfield(cuts2fit400,'bg_par_010off000')
```

```
%     if numel(cuts_list) == 3
%         bg_par =
{cuts2fit400.bg_par_010off000.p,cuts2fit400.bg_par_010off100.p,cuts2fit400.bg_par
_010off200.p};
%     else
%         bg_par  = {cuts2fit400.bg_par_010off100.p};
%     end
%     cuts_list{end+1} = bg_par;
% end
bg_par = sel_info.bg_par;
cuts_list{end+1} = bg_par;
fbase_name  = sel_info.fname;
```

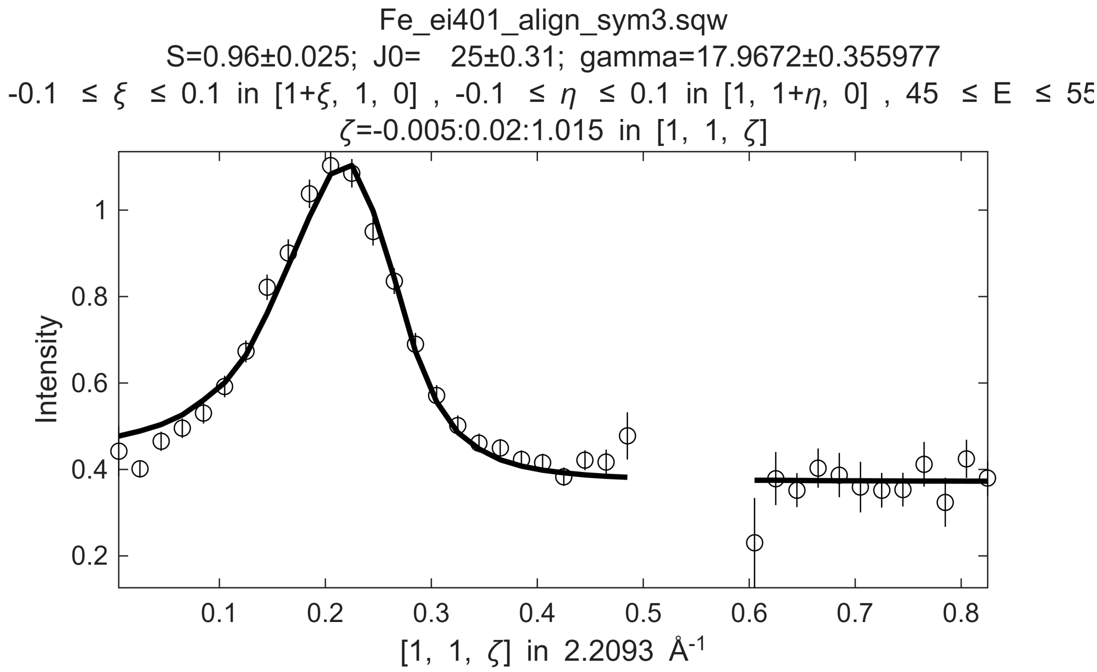
```
[fit_res_400rec,res_name] = fit_multicuts_along_direction(...
    cuts_list,fbase_name,dir_name,cut_dimensions,cut_en,dE_step,half_dE,true);
```

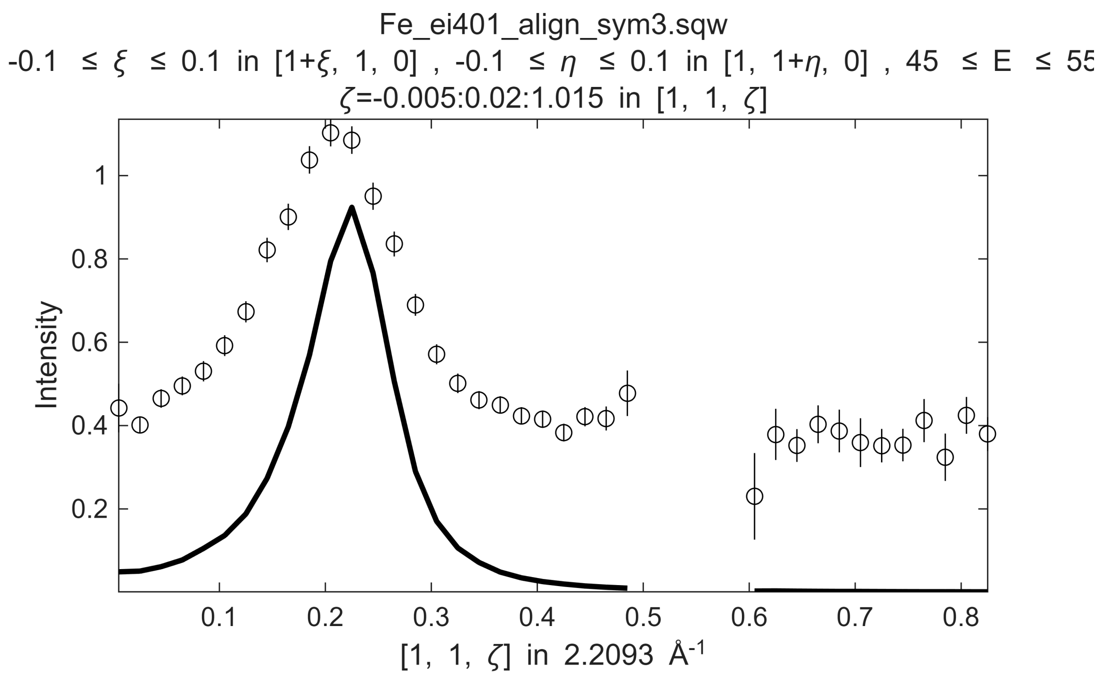



J0,
EnFit_Ei400ref001off110_1D_constFFCut_dirGH_dE10_fitSlopeDE.mat
J0 (meV)
Energy Transfer (meV)

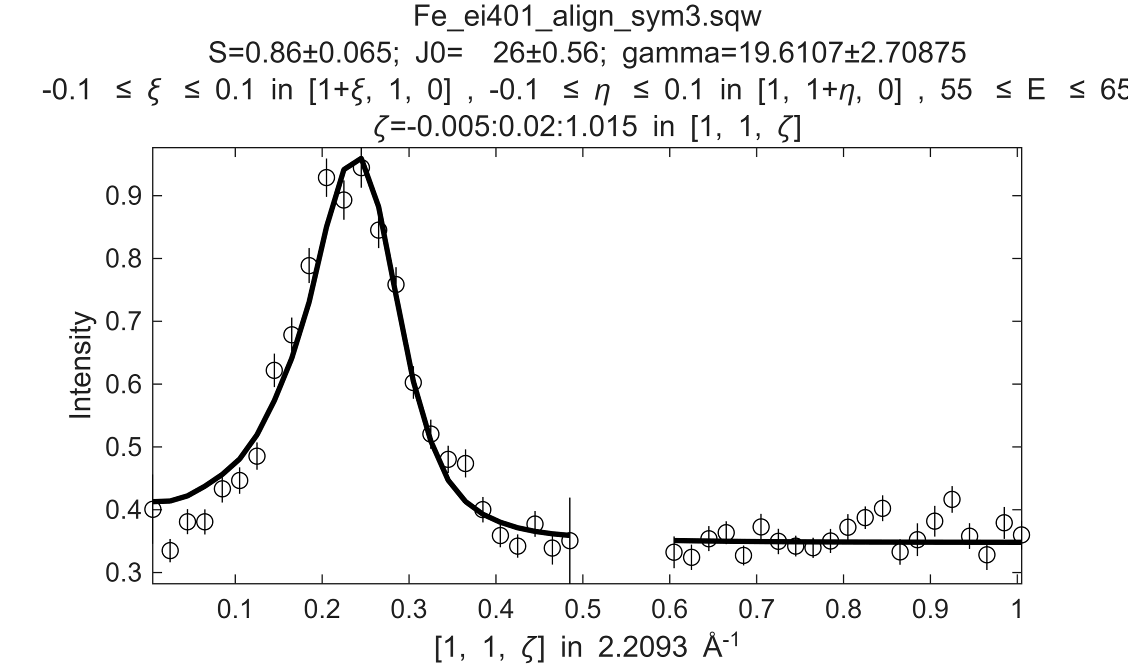

Gamma,
EnFit_Ei400ref001off110_1D_constFFCut_dirGH_dE10_fitSlopeDE.mat
DSHO broadening (meV)
Energy Transfer (meV)

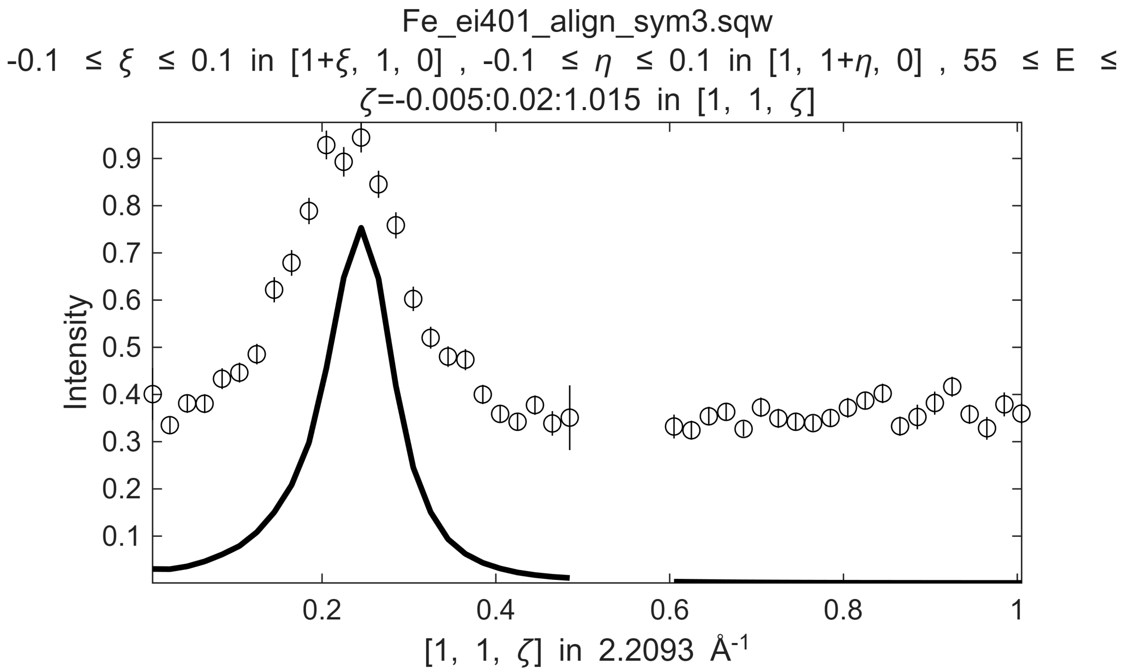
```
******************************
**** En = 50±5
******************************
```


Fe_ei401_align_sym3.sqw
S=0.96±0.025; J0= 25±0.31; gamma=17.9672±0.355977
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 45 ≤ E ≤ 55
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1


Fe_ei401_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 45 ≤ E ≤ 55
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

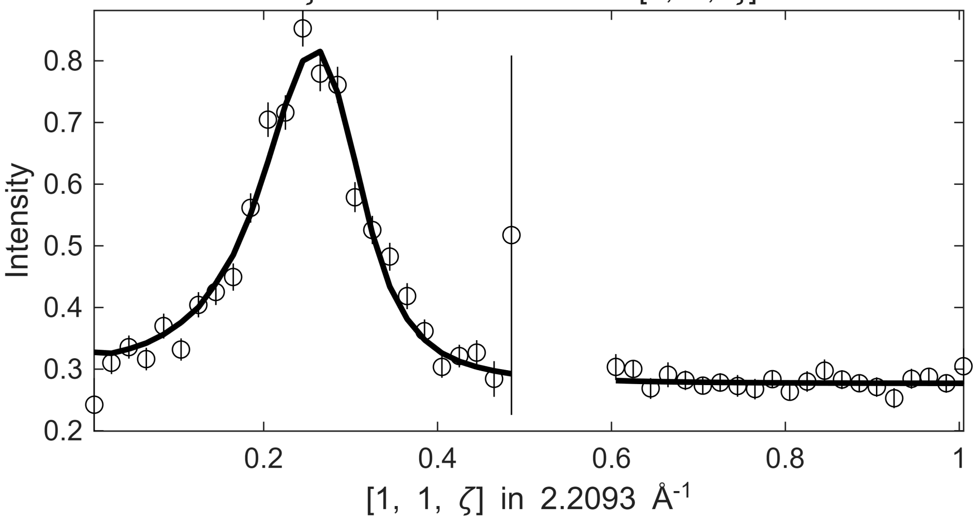
```
******************************
**** En = 60±5
******************************
```


Fe_ei401_align_sym3.sqw
S=0.86±0.065; J0= 26±0.56; gamma=19.6107±2.70875
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 55 ≤ E ≤ 65
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

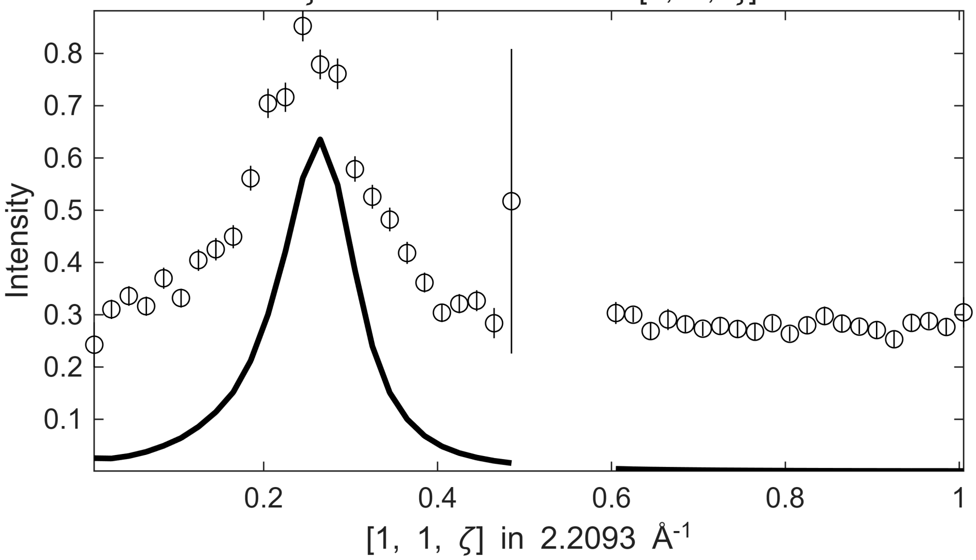

Fe_ei401_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 55 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

```
******************************
**** En = 70±5
******************************
```

Fe_ei401_align_sym3.sqw
S=0.91±0.019; J0= 27± 0.3; gamma=25.135±0.473829
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 65 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]

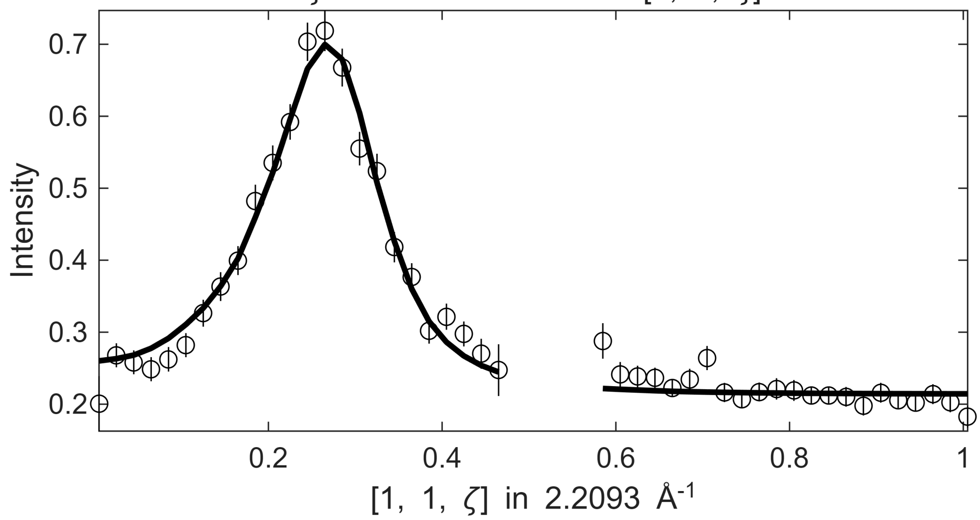


Fe_ei401_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 65 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]

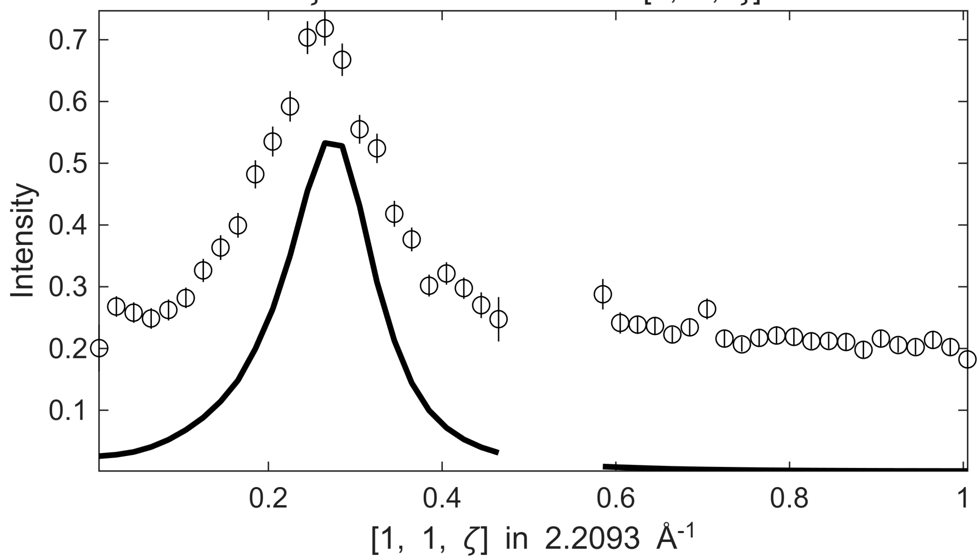



```
******************************
**** En = 80±5
******************************
```

Fe_ei401_align_sym3.sqw
S= 1.1±0.024; J0= 29±0.36; gamma=36.1174±0.670484
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 75 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]

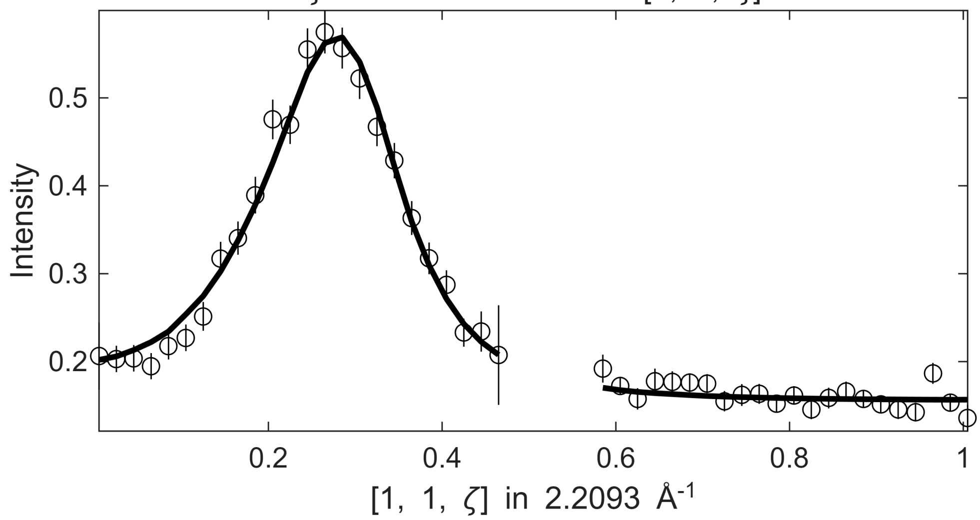


Fe_ei401_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 75 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]

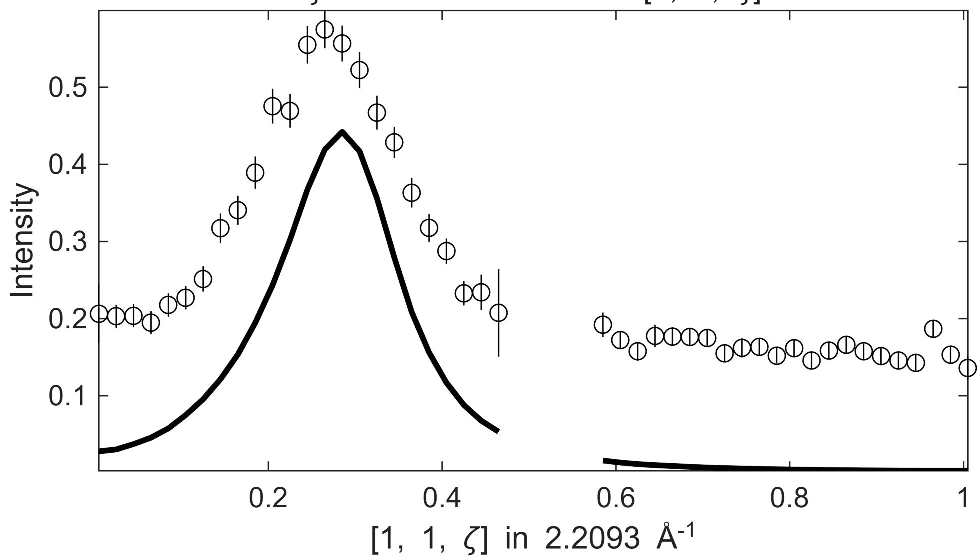



```
******************************
**** En = 90±5
******************************
```

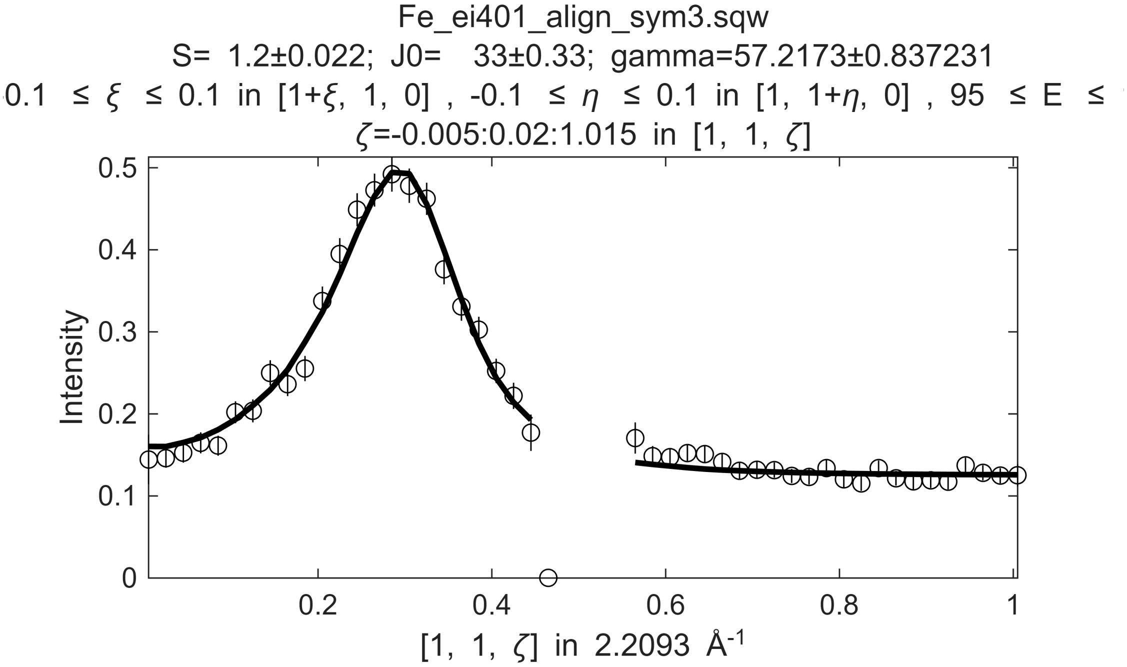
Fe_ei401_align_sym3.sqw
S= 1.4±0.026; J0= 33±0.37; gamma=62.6609±0.923968
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 85 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]

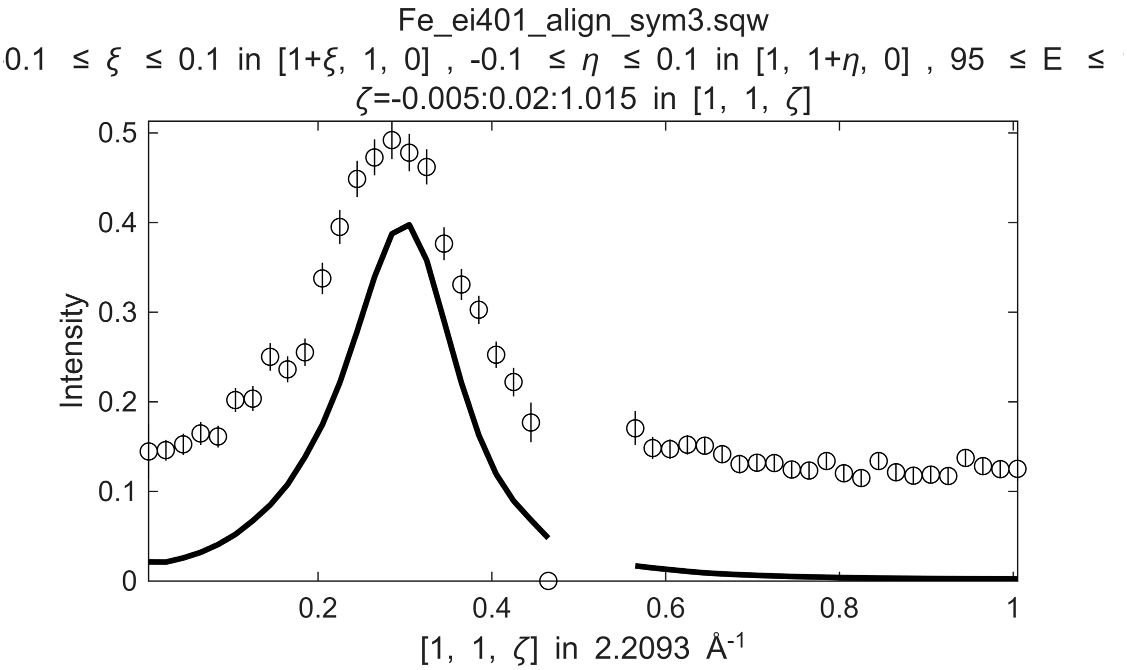


Fe_ei401_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 85 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]

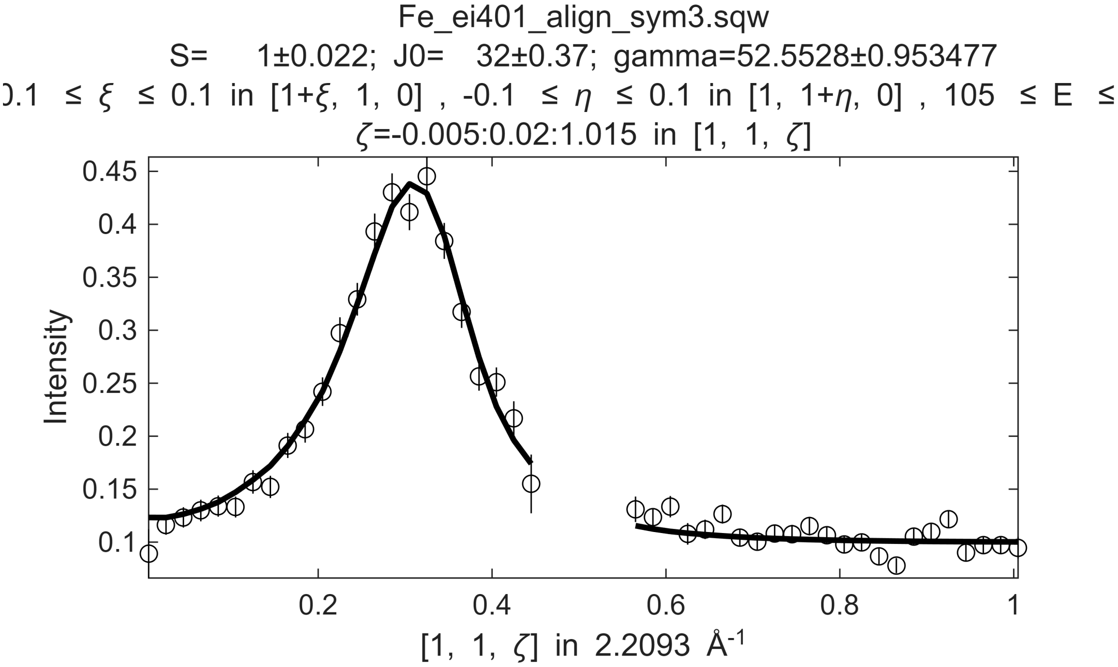



```
******************************
**** En = 100±5
******************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.9608
```

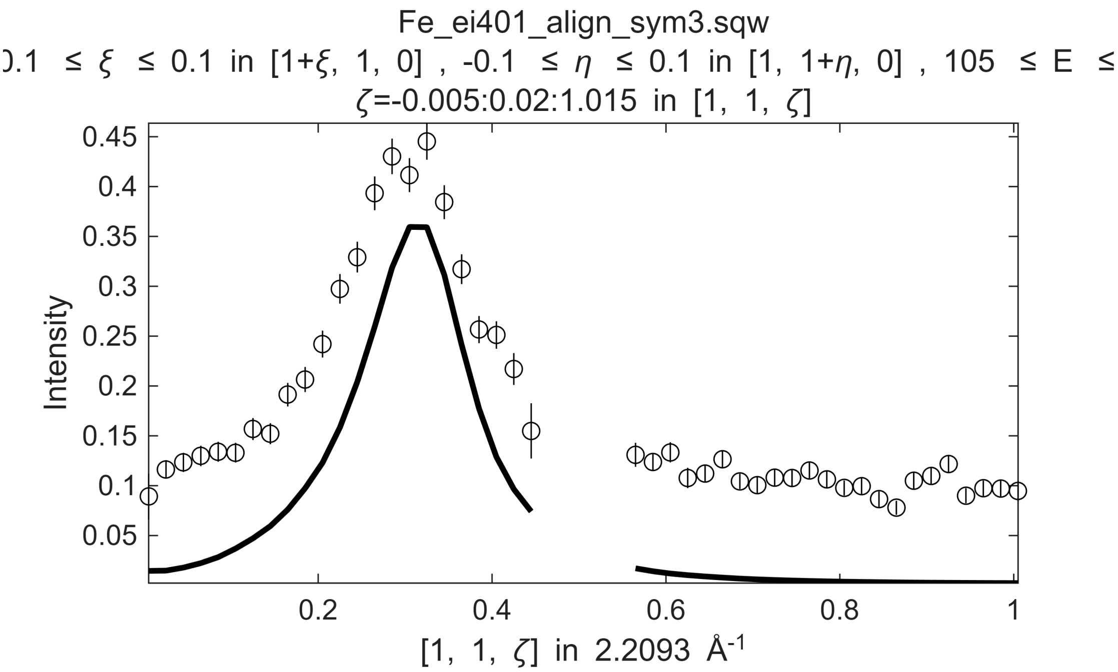

Fe_ei401_align_sym3.sqw
S= 1.2±0.022; J0= 33±0.33; gamma=57.2173±0.837231
0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 95 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
0.5
0.4
0.3
0.2
0.1
0
0.2
0.4
0.6
0.8
1
[1, 1, ζ] in 2.2093 Å-1


Fe_ei401_align_sym3.sqw
0.1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 95 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
0.5
0.4
0.3
0.2
0.1
0
0.2
0.4
0.6
0.8
1
[1, 1, ζ] in 2.2093 Å-1

```
******************************
**** En = 110±5
******************************
```

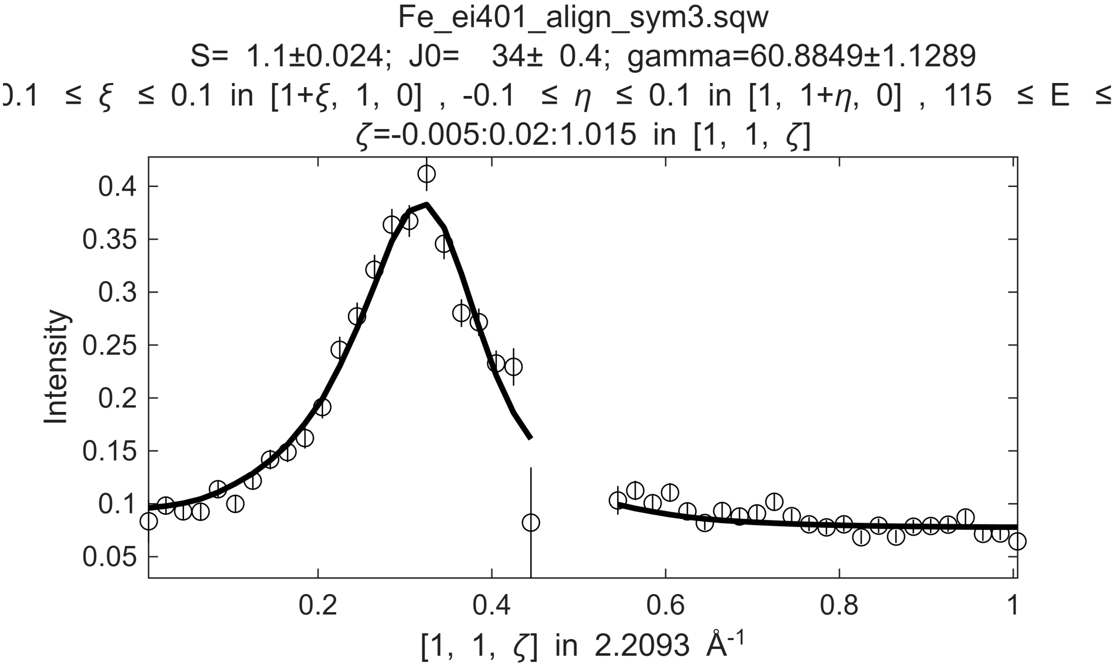


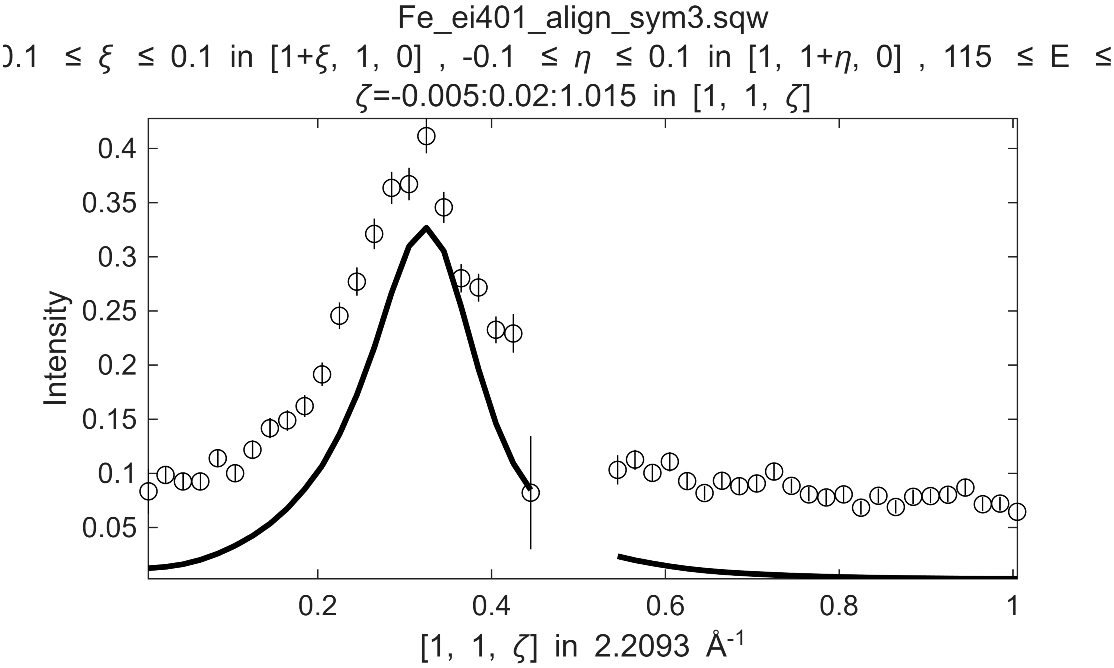
Fe_ei401_align_sym3.sqw
S= 1±0.022; J0= 32±0.37; gamma=52.5528±0.953477
).1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 105 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1



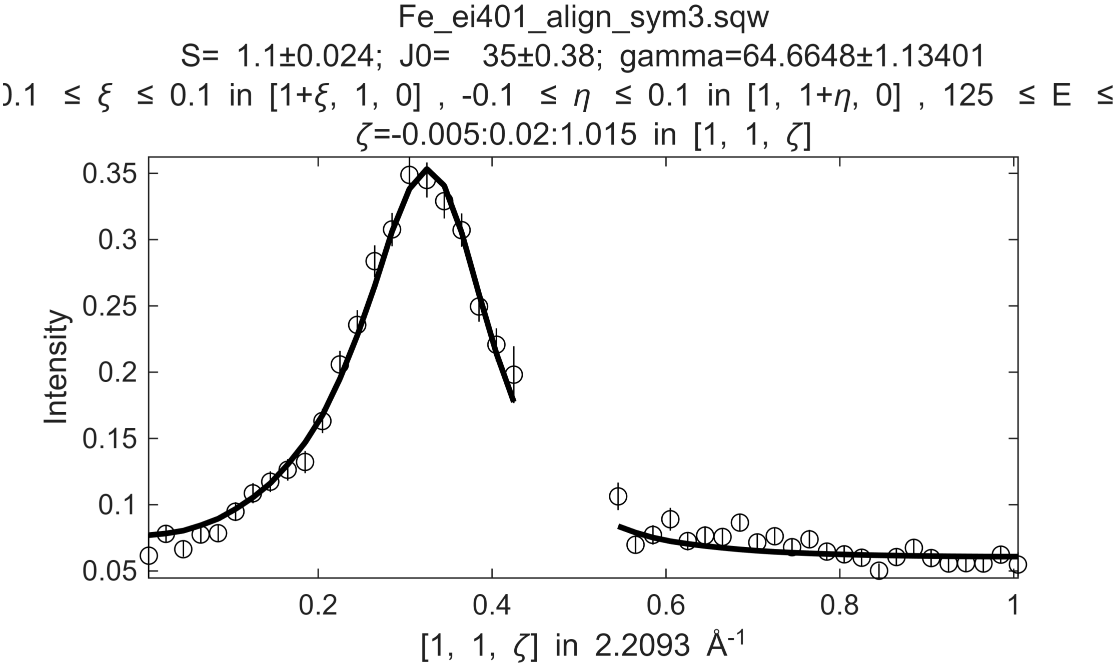
Fe_ei401_align_sym3.sqw
).1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 105 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

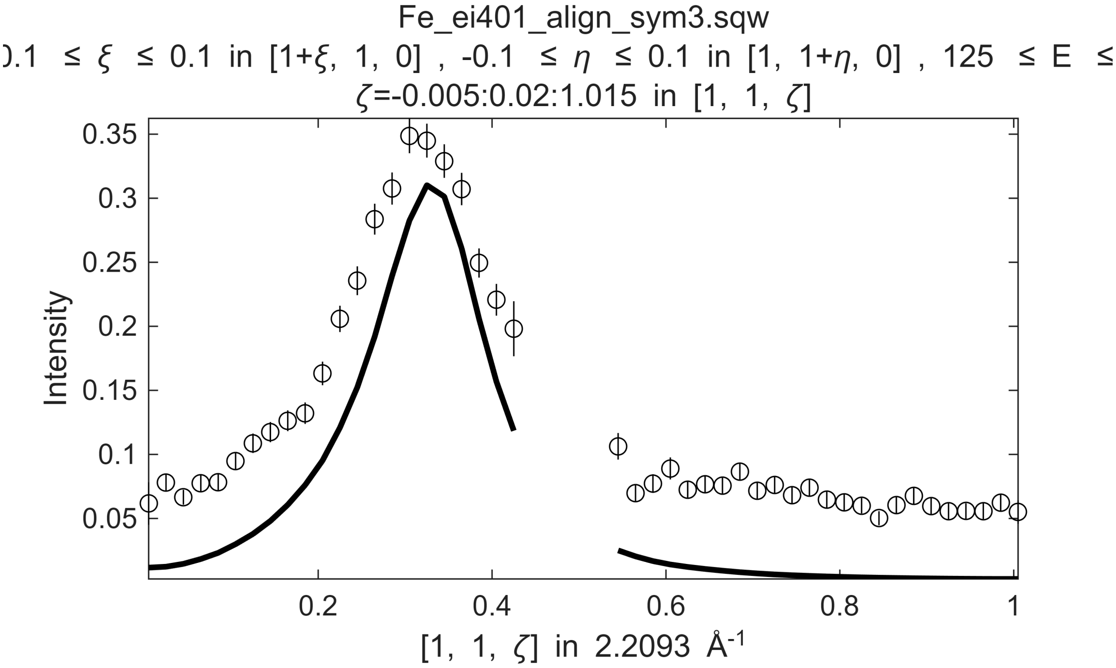

```
******************************
**** En = 120±5
******************************
```


Fe_ei401_align_sym3.sqw
S= 1.1±0.024; J0= 34± 0.4; gamma=60.8849±1.1289
).1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 115 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

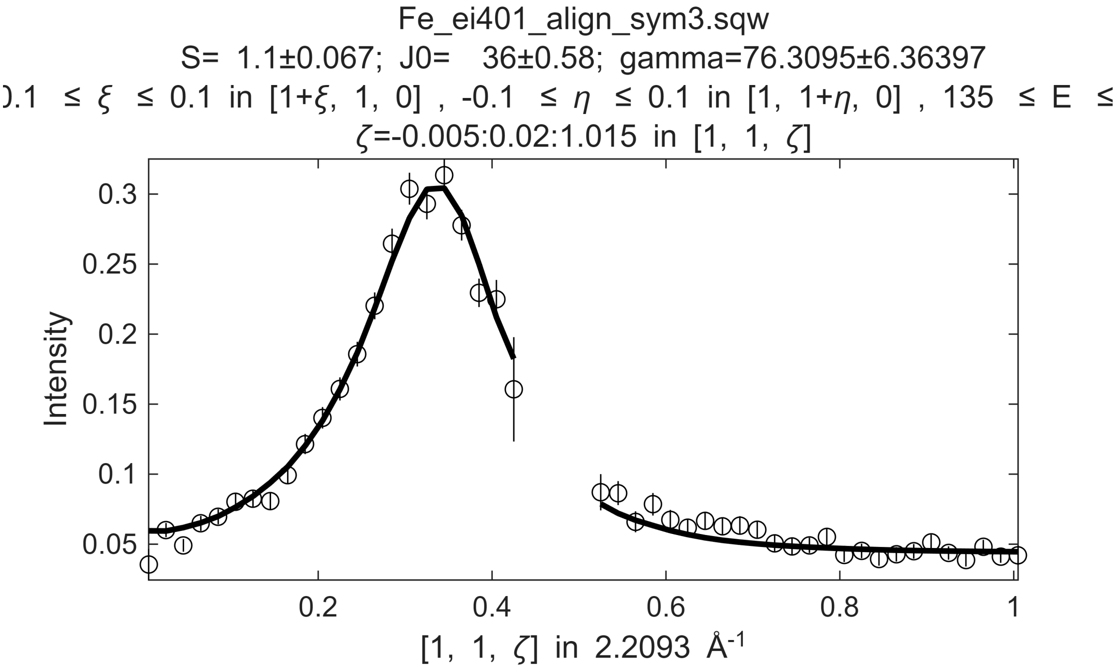

Fe_ei401_align_sym3.sqw
).1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 115 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

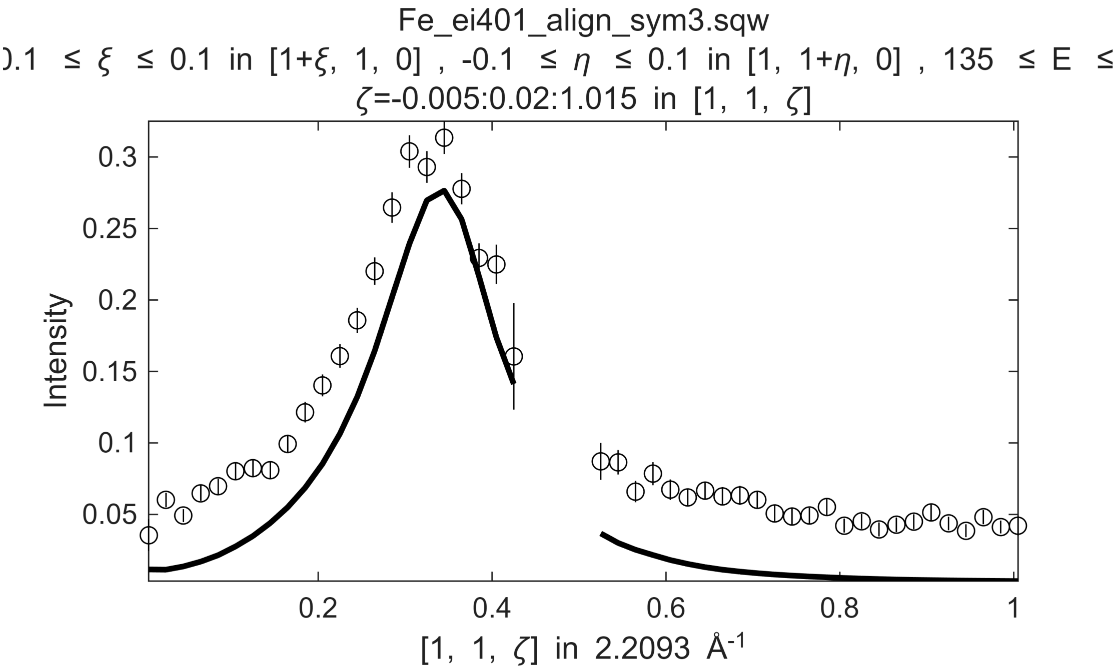
```
******************************
**** En = 130±5
******************************
```


Fe_ei401_align_sym3.sqw
S= 1.1±0.024; J0= 35±0.38; gamma=64.6648±1.13401
).1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 125 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

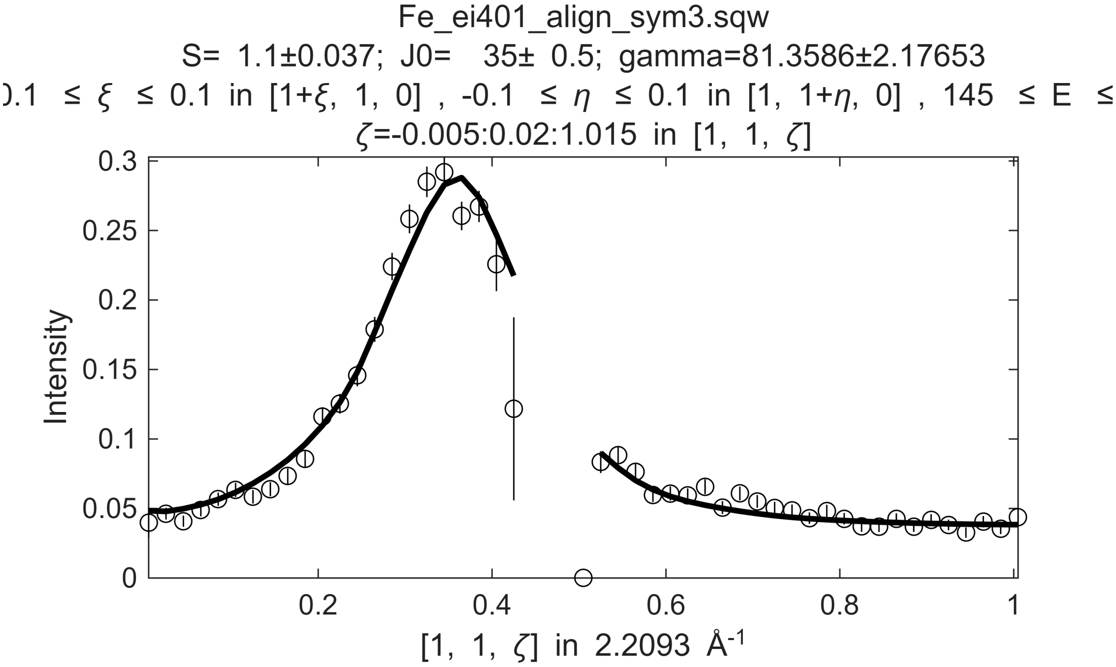

Fe_ei401_align_sym3.sqw
).1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 125 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

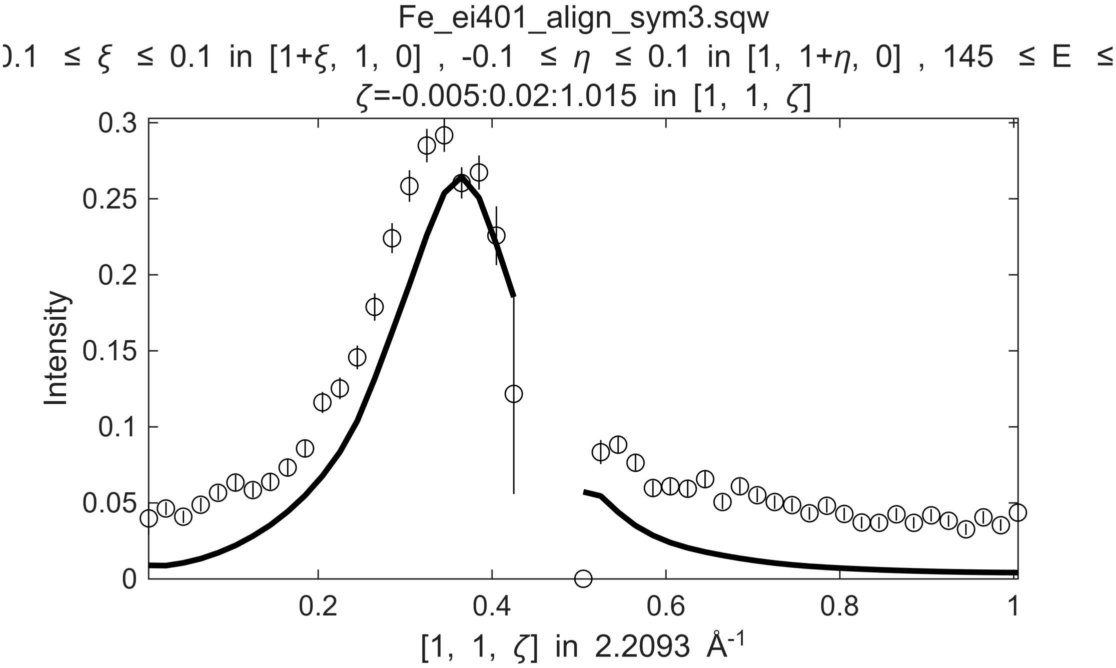
****
**** En = 140±5
****



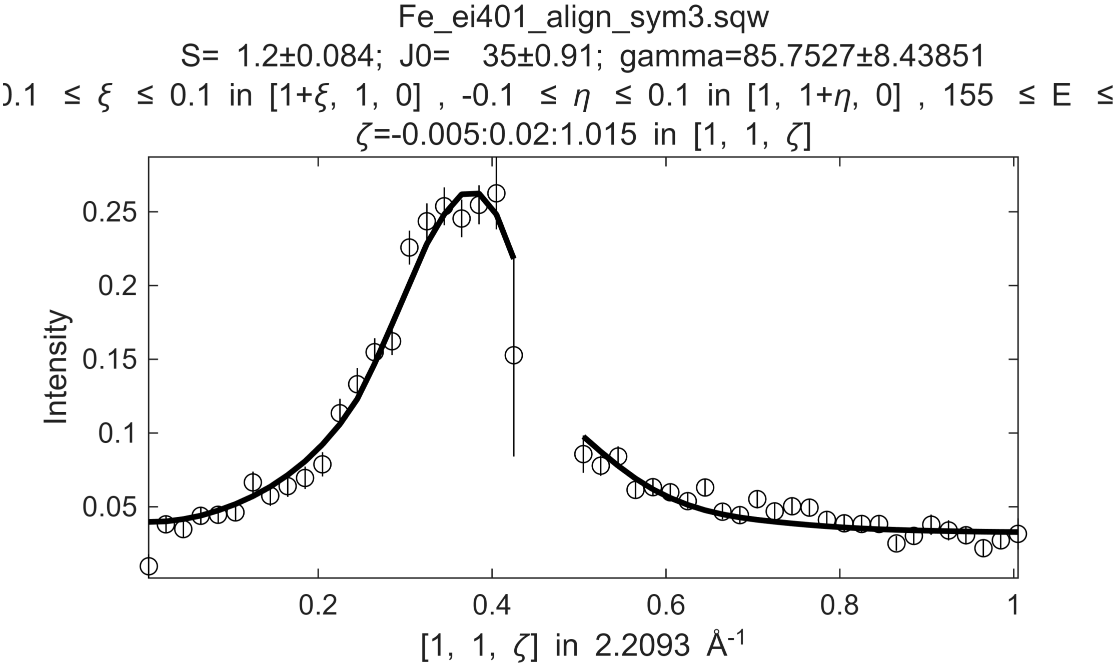


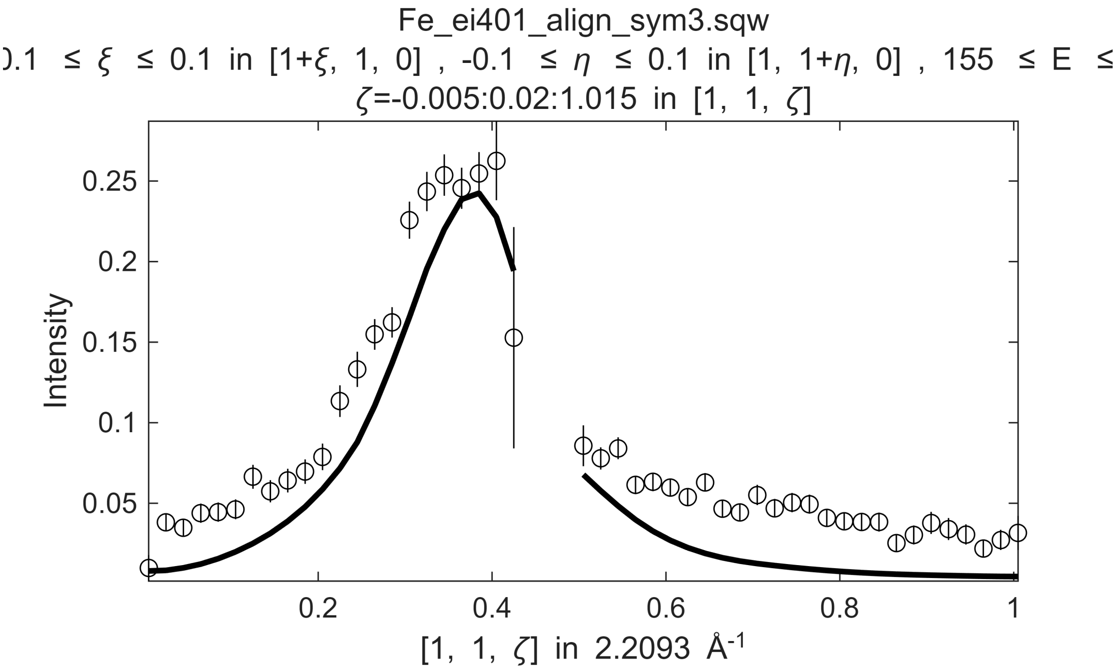

```
******************************
**** En = 150±5
******************************
Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed
from fit, which is 1.9608
```

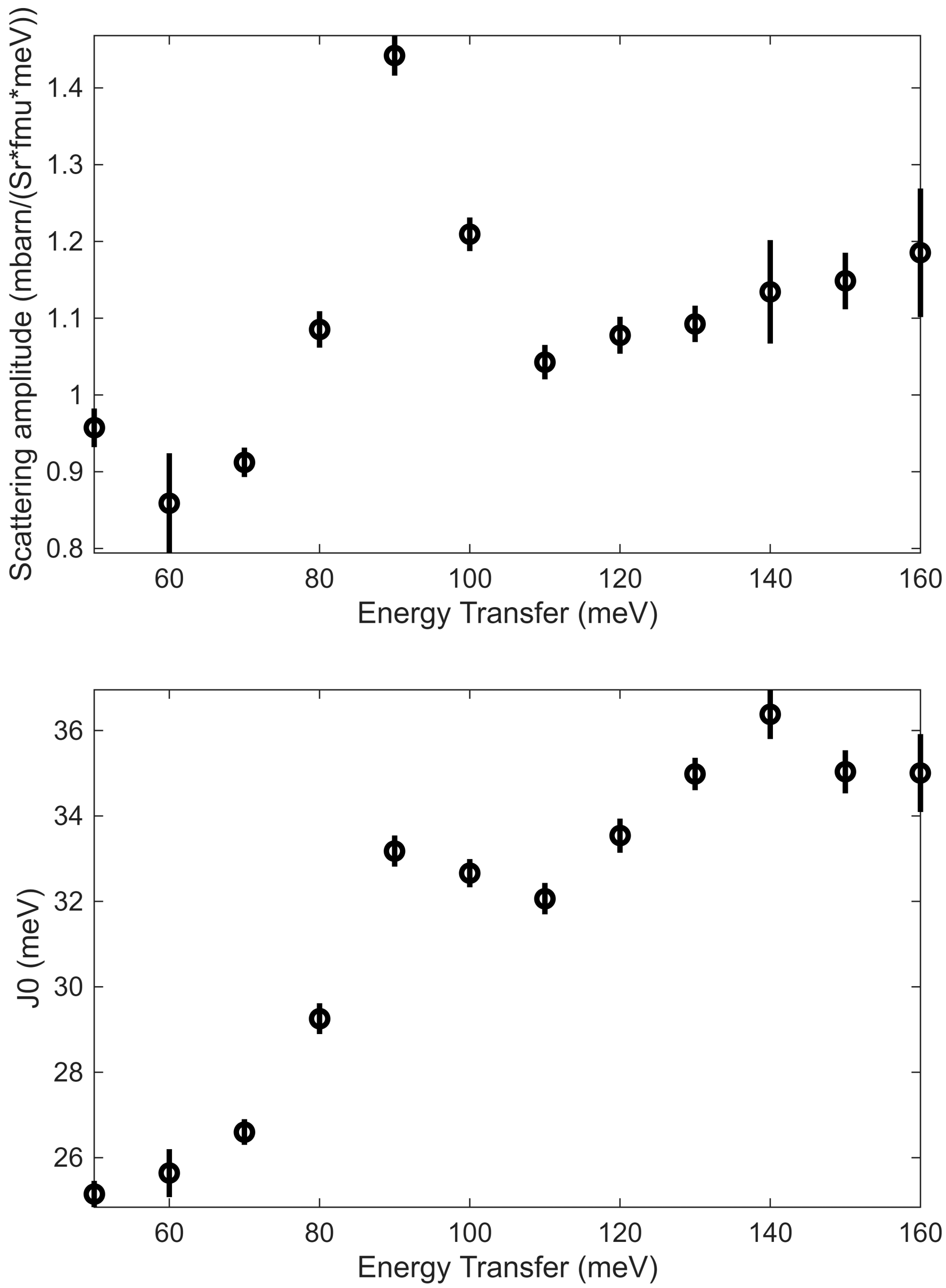

Fe_ei401_align_sym3.sqw
S= 1.1±0.037; J0= 35± 0.5; gamma=81.3586±2.17653
).1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 145 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

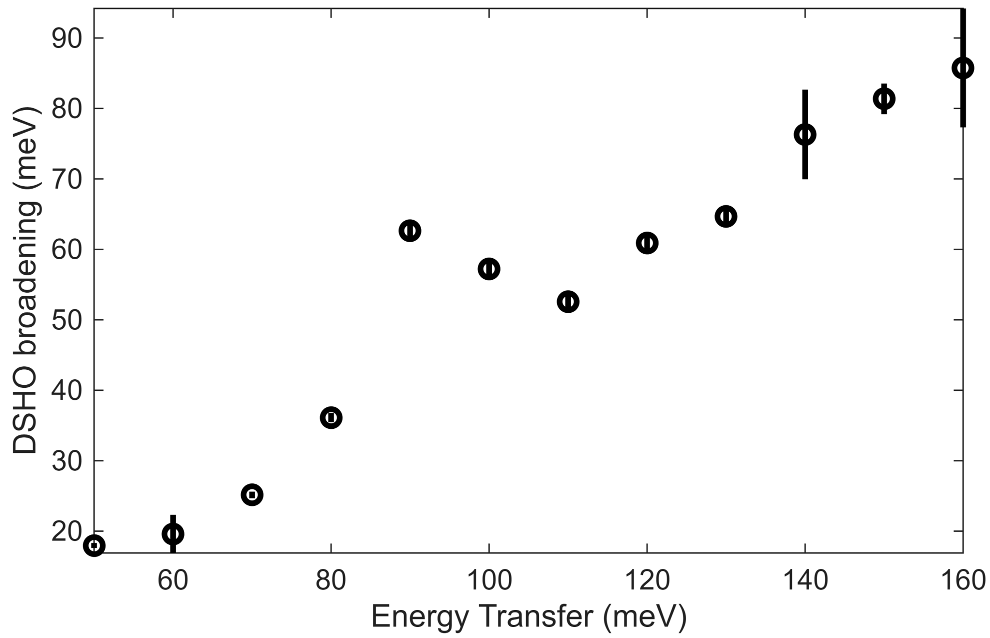

Fe_ei401_align_sym3.sqw
).1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 145 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

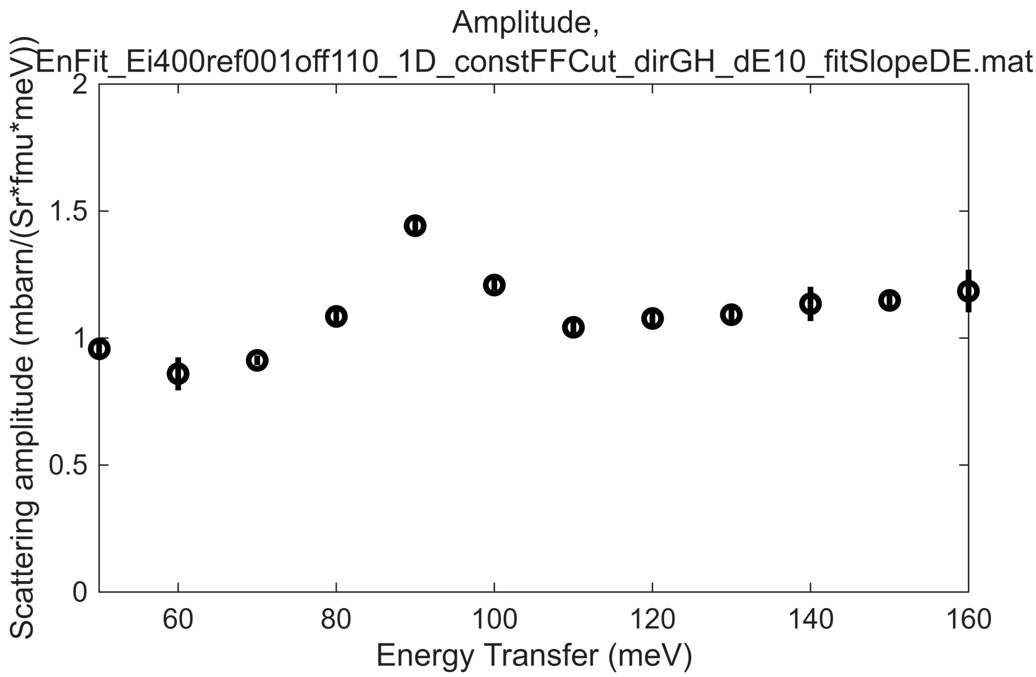
```
******************************
**** En = 160±5
******************************
```

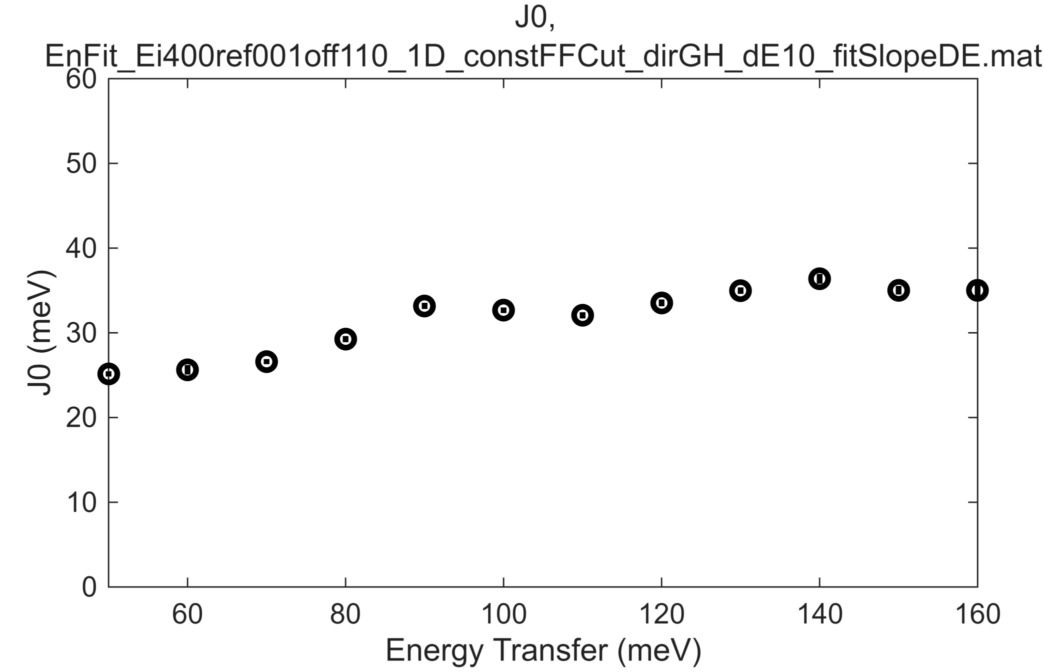

Fe_ei401_align_sym3.sqw
S= 1.2±0.084; J0= 35±0.91; gamma=85.7527±8.43851
).1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 155 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

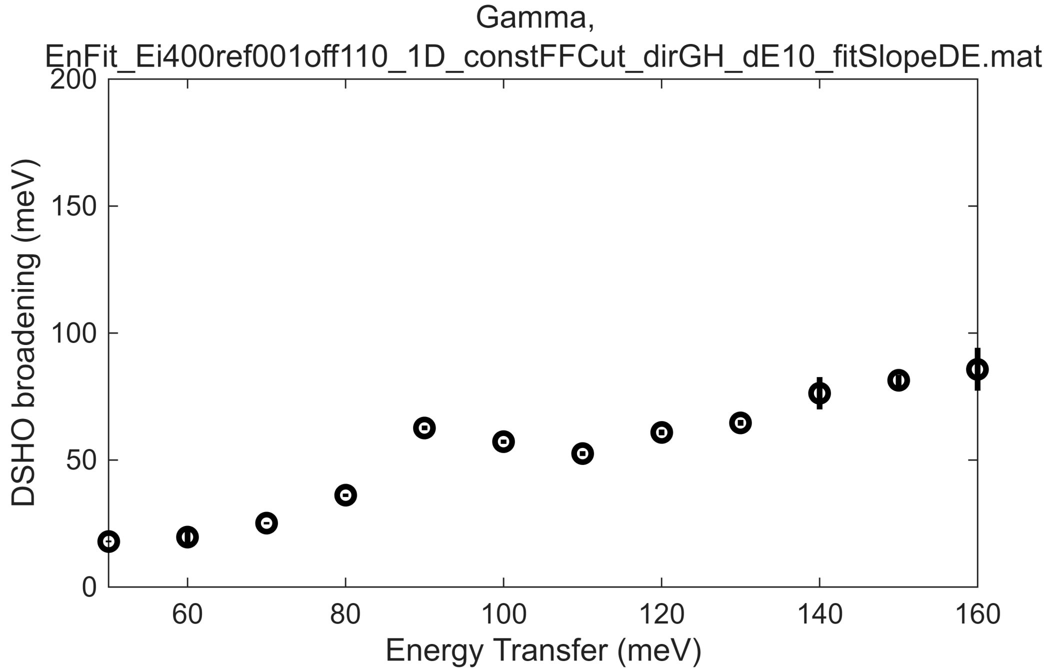

Fe_ei401_align_sym3.sqw
).1 ≤ ξ ≤ 0.1 in [1+ξ, 1, 0] , -0.1 ≤ η ≤ 0.1 in [1, 1+η, 0] , 155 ≤ E ≤
ζ=-0.005:0.02:1.015 in [1, 1, ζ]
Intensity
[1, 1, ζ] in 2.2093 Å-1

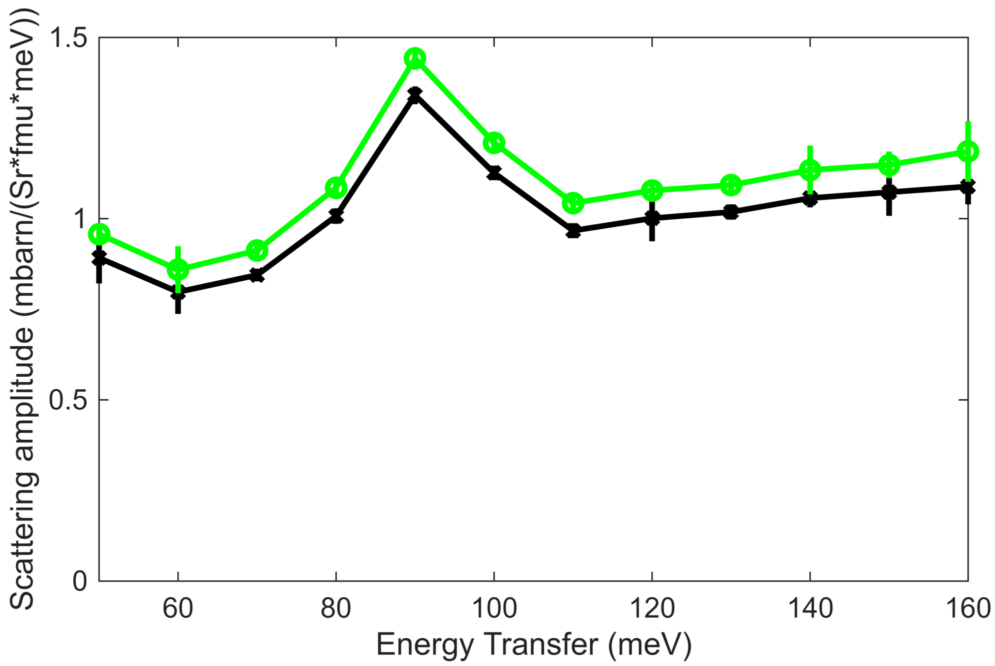

Scattering amplitude (mbarn/(Sr*fmu*meV))
Energy Transfer (meV)
J0 (meV)
Energy Transfer (meV)

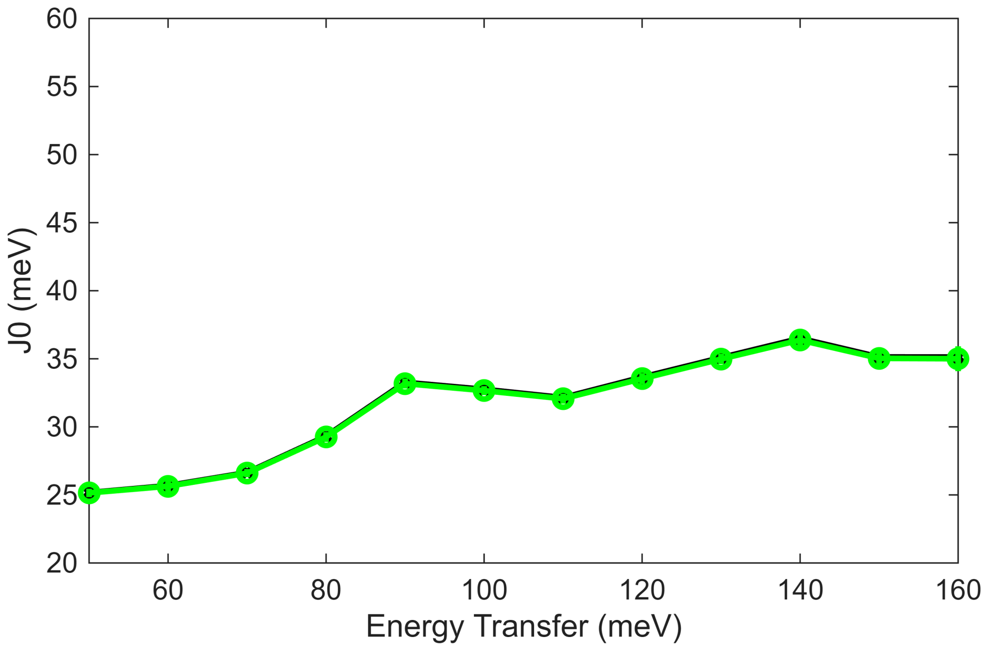

DSHO broadening (meV)
Energy Transfer (meV)
20
30
40
50
60
70
80
90
60
80
100
120
140
160

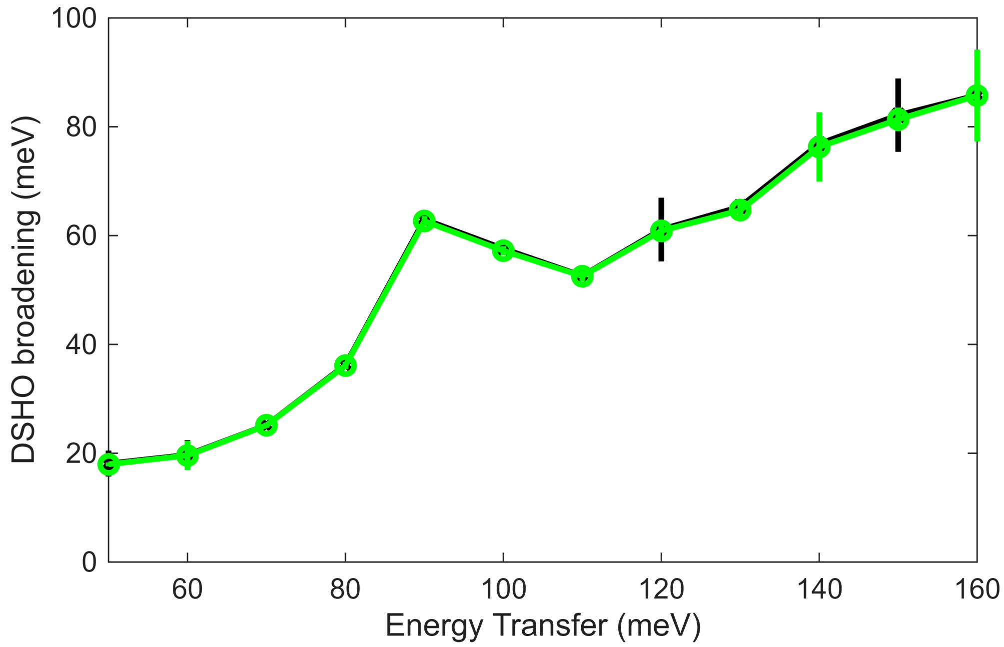

Amplitude,
EnFit_Ei400ref001off110_1D_constFFCut_dirGH_dE10_fitSlopeDE.mat
Scattering amplitude (mbarn/(Sr*fmu*meV))
Energy Transfer (meV)
0
0.5
1
1.5
2
60
80
100
120
140
160


J0,
EnFit_Ei400ref001off110_1D_constFFCut_dirGH_dE10_fitSlopeDE.mat
J0 (meV)
0
10
20
30
40
50
60
60
80
100
120
140
160
Energy Transfer (meV)


Gamma,
EnFit_Ei400ref001off110_1D_constFFCut_dirGH_dE10_fitSlopeDE.mat
DSHO broadening (meV)
0
50
100
150
200
60
80
100
120
140
160
Energy Transfer (meV)

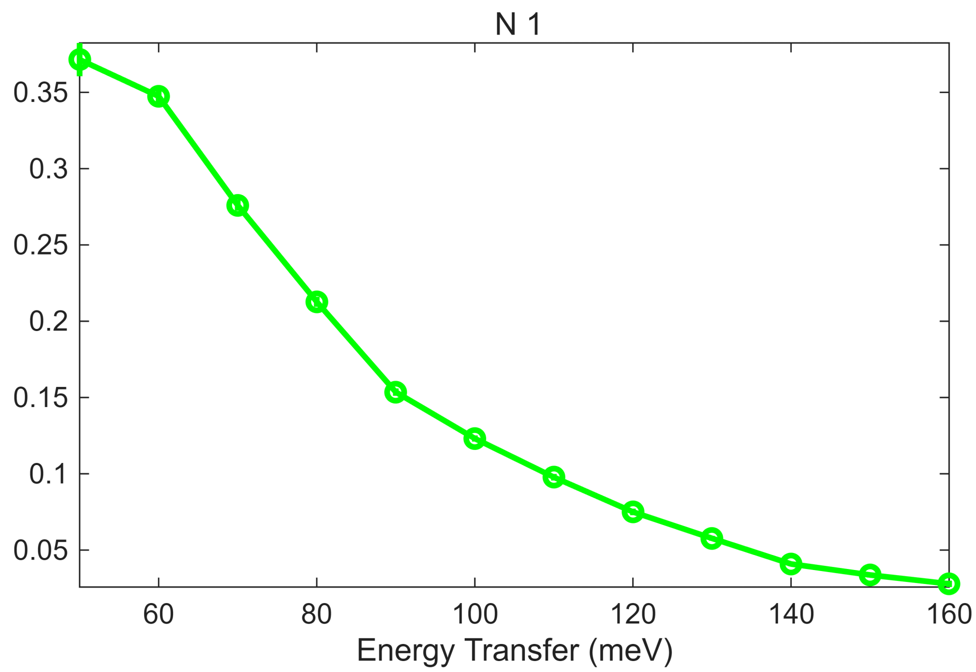

Scattering amplitude (mbarn/(Sr*fmu*meV))
1.5
1
0.5
0
60
80
100
120
140
160
Energy Transfer (meV)


J0 (meV)
60
55
50
45
40
35
30
25
20
60
80
100
120
140
160
Energy Transfer (meV)

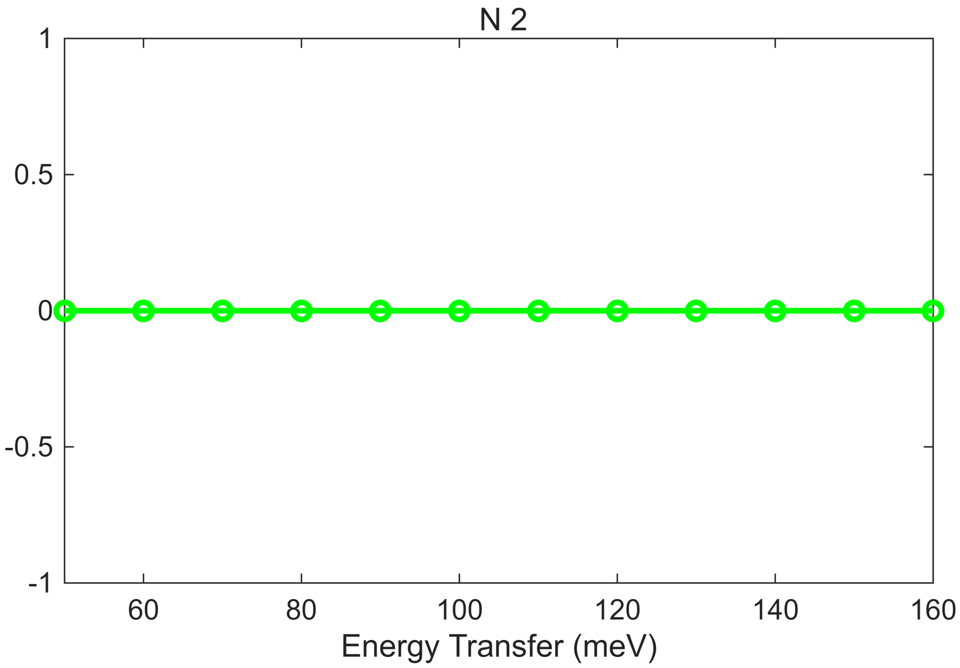



```
%cl,'En_cuts800',dir_name,cut_en,dE_step,half_dE);
beep
beep
beep
```

```
fit_res_400 = fit_res_400rec;

bg_data = extract_bg_par(fit_res_400.all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(fit_res_400.all_fit_par(1).en,fit_res_400.all_fit_par(end).en);
keep_figure;
```



```
pd(bg_data(2));lx(fit_res_400.all_fit_par(1).en,fit_res_400.all_fit_par(end).en);
keep_figure
```




```
fit_resEi400 = fit_res_400rec;
save(res_name,"fit_resEi400")
```